

Lesson 13: Quantum Chemistry with VQE

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H₂, Hamiltonians, Jordan–Wigner mapping, ansatz, and variational optimization

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Learning Outcomes

- Explain the idea of ground-state energy.
- Describe the chemistry-to-qubit pipeline.
- Understand the role of Jordan–Wigner mapping and VQE.
- Interpret the reported H_2 energy.

Why Quantum Computing for Chemistry?

- Molecules are quantum systems.
- Their electronic structure can be represented by Hamiltonians.
- The goal in this notebook is to estimate the ground-state energy of H_2 .
- The workflow transforms a chemistry problem into a qubit Hamiltonian and then optimizes a parameterized circuit.

H₂ in the Notebook

- Geometry: two hydrogen atoms separated by 0.735 Å.
- Basis: STO-3G.
- PySCF supplies molecular-orbital/electronic-structure calculations.
- Qiskit Nature manages the chemistry representation.

Molecule $\rightarrow$ Electronic Structure $\rightarrow$ Qubit Hamiltonian $\rightarrow$ VQE $\rightarrow$ Energy

Read the Main Chemistry Code

```
molecule_geometry = "H 0 0 0; H 0 0 0.735"  
driver = PySCFDriver(atom=molecule_geometry, basis="sto3g")  
es_problem = ElectronicStructureProblem(driver)  
transformer = FreezeCoreTransformer()  
es_problem = transformer.transform(es_problem)  
mapper = JordanWignerMapper()  
ansatz = TwoLocal(rotation_blocks=["ry", "rz"], entanglement_blocks="cz")  
optimizer = COBYLA(maxiter=100)  
estimator = Estimator()  
vqe = VQE(estimator, ansatz, optimizer)  
hamiltonian = mapper.map(es_problem.hamiltonian.second_q_op())  
result = vqe.compute_minimum_eigenvalue(hamiltonian)  
print(f"Estimated ground state energy: {result.optimal_value:.4f} Hartree")
```

- PySCFDriver creates the electronic-structure problem.
- FreezeCoreTransformer applies the notebook's optional approximation.
- JordanWignerMapper converts fermionic operators into qubit operators.
- TwoLocal supplies a parameterized ansatz.
- VQE optimizes the expectation value of the Hamiltonian.

What Is VQE?

- ① Prepare a parameterized trial state.
- ② Evaluate the Hamiltonian expectation value.
- ③ Use a classical optimizer to update the circuit parameters.
- ④ Repeat until the energy is minimized.

Key idea: the quantum circuit proposes states; the classical optimizer searches the parameter space.

Estimated ground state energy: -1.1373 Hartree

- The optimizer found a low-energy variational state for the model Hamiltonian used by the notebook.
- The negative sign means the computed energy lies below the chosen zero-energy reference.
- The value is an estimate produced by the specified electronic-structure model and VQE configuration.

Beginner Activity

- Change the H–H distance and discuss why molecular energy can change.
- Change the optimizer iteration limit.
- Discuss why chemistry problems require both a physical model and a numerical optimizer.

- Quantum chemistry can be expressed as qubit Hamiltonians.
- Jordan–Wigner maps fermionic operators to qubit operators.
- VQE searches for a low-energy quantum state.
- The notebook estimates H_2 ground-state energy as -1.1373 Hartree.

VQE is a central example of hybrid quantum-classical optimization for scientific computing.

Quick Check

- ① What is the ground state?
- ② Why do we need a Hamiltonian?
- ③ What does VQE optimize?
- ④ What role does Jordan–Wigner mapping play?